INCH-POUND

MIL-PRF-8805/1E
31 October 2001
SUPERSEDING
MIL-PRF-8805/1D
24 September 1969

PERFORMANCE SPECIFICATION SHEET
SWITCHES, SENSITIVE, SPDT, UNSEALED

This specification forms a part of MIL-PRF-8805, dated 23 January 1998, and is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and MIL-PRF-8805.

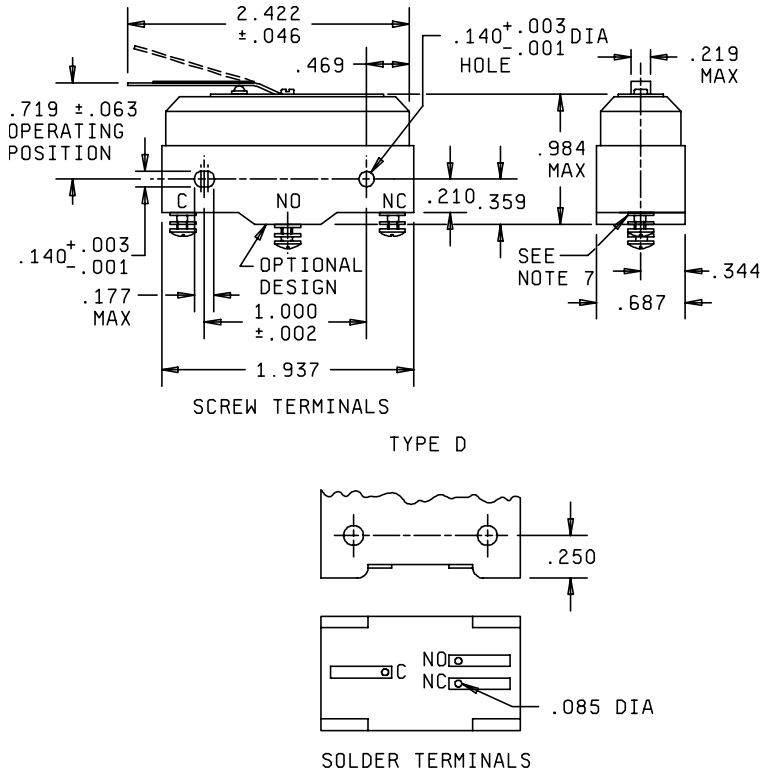


FIGURE 1. Dimensions and configurations.

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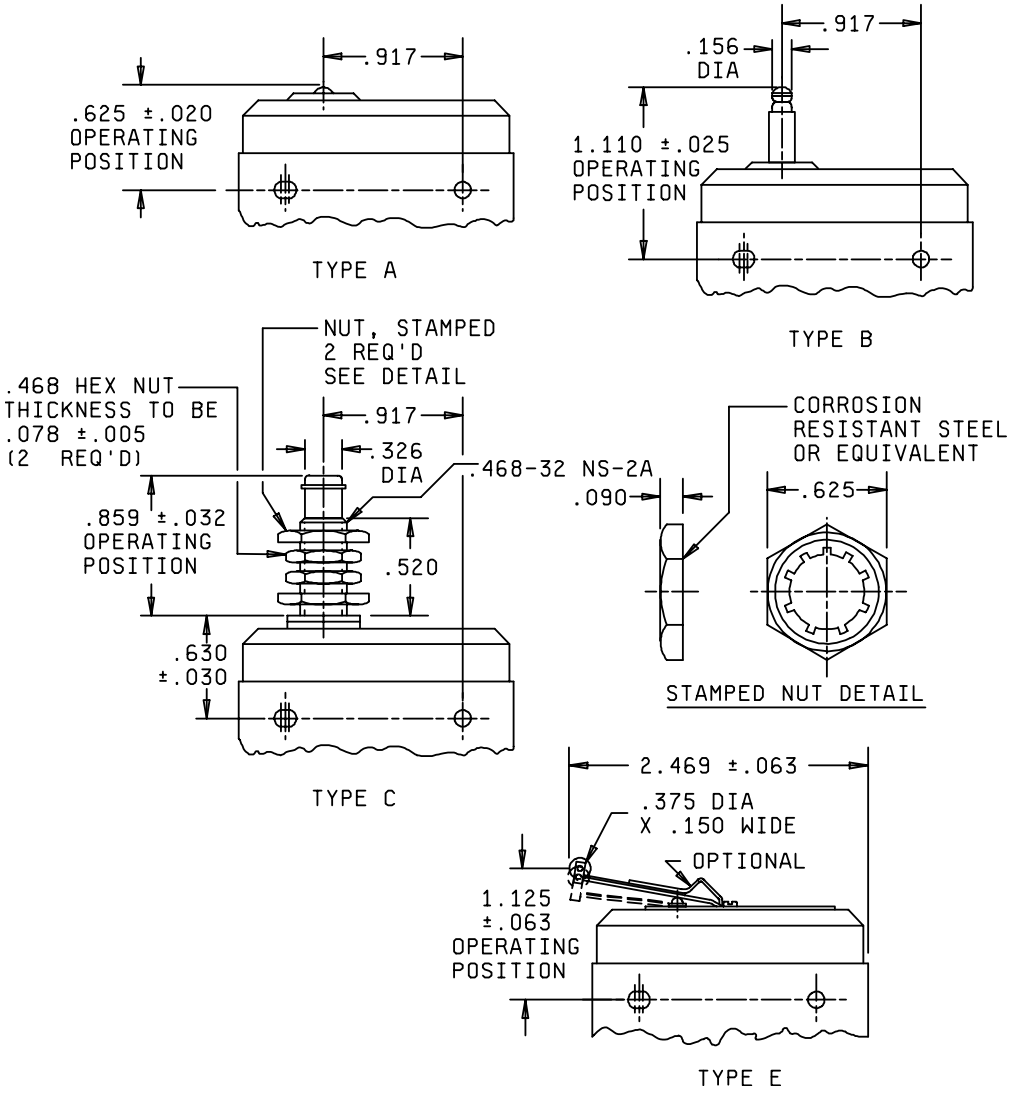
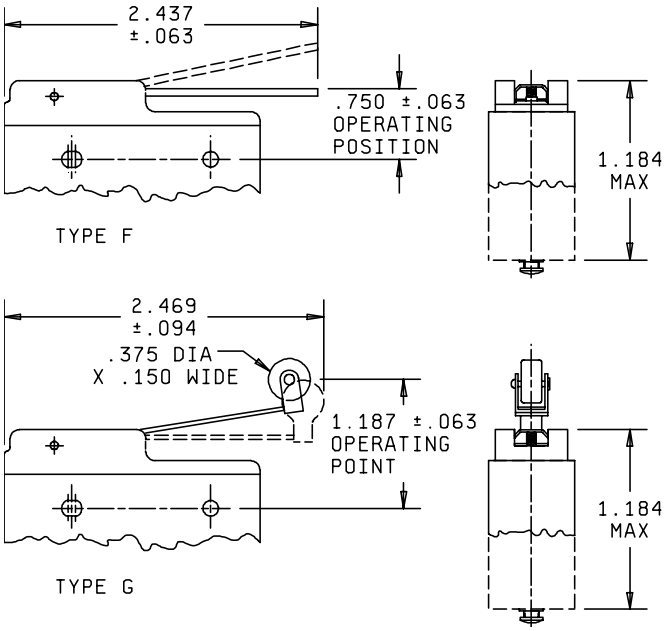


FIGURE 1. Dimensions and configurations -Continued.

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Inches	mm	Inches	mm	Inches	mm	Inches	mm	Inches	mm	Inches	mm
.001	.03	.063	1.60	.177	4.50	.375	9.53	.750	19.05	1.184	30.07
.002	.05	.085	2.16	.210	5.33	.469	11.91	.859	21.82	1.187	30.15
.003	.08	.090	2.29	.219	5.56	.520	13.21	.917	23.29	1.937	49.20
.020	.51	.094	2.39	.250	6.35	.625	15.88	.984	24.99	2.422	61.52
.025	.64	.140	3.56	.326	8.28	.630	16.00	1.000	25.40	2.437	61.90
.032	.81	.150	3.81	.344	8.74	.687	17.45	1.110	28.19	2.469	62.71
.046	1.17	.156	3.96	.359	9.12	.719	18.26	1.125	28.58		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise stated, tolerance is ±.015 (0.38 mm).
4. Exact shape of switch optional provided dimensions specified are not exceeded.
5. Terminals: Captive construction or equivalent.
   a. Screw terminals are supplied with lockwashers and No. 6-32 NC-2A, Fil HD, binder HD, or pan HD screws.
   b. Solder lug terminals shall accommodate one No. 14 AWG wire.
6. Mounting holes shall accept .139 (3.53 mm) maximum diameter pins or screws on 1.000 ±.002 (25.40 ±.05 mm) centers.
7. Switch terminal surface shall be free of plastic flash and may extend .030 (.76 mm) maximum beyond plastic body surface.
8. Actuators may be permanently affixed by threading into the plastic cover.

FIGURE 1. Dimensions and configurations -Continued.

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REQUIREMENTS:

Dimensions and configurations: See figure 1.

Enclosure design: 1 (unsealed).

Temperature characteristic: 1 (-55°C to +85°C).

Shock type: M (100 g), (contacts not monitored during test).

Vibration grade: 1 (10 to 500 Hz) (momentary transfer of contacts permitted in free position; no contact transfer, in excess of 10 microseconds, permitted in actuated position).

Weight: .1 pound maximum.

Operating characteristics: See table 1.

Strength of actuating means:

Type A and B:	10 pounds.
Type C:	25 pounds.
Type D, E, F, and G:	8 ounces

Terminal strength: (solder terminals): 5 pounds.

Explosion: Not applicable.

Contact resistance: Not applicable.

Acceleration: 50g (except type E and type G shall be 20g).

Moisture resistance: Minimum insulation resistance, wet condition, 1 megohm.

Dielectric withstanding voltage:
Sea level: 600 V rms between all terminals and ground.
Altitude: 300 V rms at 50,000 ft.
(In qualification inspection table after electrical endurance the dielectric withstanding voltage points of application between all unconnected terminals of the same pole is not applicable).

Mechanical endurance: See table II.

Electrical ratings: See table II.

Current ratings: Application information (see table IV and V).

Qualification:
Group submission: See table III.

Part or Identifying Number (PIN): See table VI.

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Table I. Operating characteristics and PIN's. 1/

Type		Nominal contact separation (See table IV.) (Inch) 3/	Maximum operating force (Ounce)	Minimum release force (Ounce)	Maximum differential travel (Inch)	Travel		PIN's 2/	
						Maximum pretravel (Inch)	Minimum overtravel (Inch)	Terminal	
								Solder	Screw
	Pin plunger	.010	10	4	.003	.040	.005	-028	-027
	Pin plunger	.020	13	4	.005	.040	.005	-003	-004
A	Pin plunger	.040	20	4	.007	.040	.005	-005	-006
	Pin plunger	.070	28	4	.010	.040	.005	-007	-008
	Spring plunger	.010	10	4	.003	.040	.050	-009	-010
	Spring plunger	.020	13	4	.005	.040	.050	-011	-012
B	Spring plunger	.040	20	4	.007	.040	.050	-013	-014
	Spring plunger	.070	28	4	.010	.040	.050	-015	-016
	Panel mount ¼-inch overtravel	.010	10	4	.003	.040	.203	-017	-018
	Panel mount ¼-inch overtravel	.020	13	4	.005	.040	.203	-019	-020
C	Panel mount ¼-inch overtravel	.040	20	4	.007	.040	.203	-021	-022
	Panel mount ¼-inch overtravel	.070	28	4	.010	.040	.203	-023	-024
	Leaf spring	.010	5	.5	.030	.156	.031	-025	-026
	Leaf spring	.020	6	.5	.050	.156	.031	-002	-001
D	Leaf spring	.040	7	.5	.065	.156	.031	-029	-030
	Leaf spring	.070	9	.5	.080	.219	.031	-031	-032
	Roller leaf spring	.010	5	.5	.030	.188	.031	-033	-034
	Roller leaf spring	.020	6	.5	.050	.188	.031	-035	-036
E	Roller leaf spring	.040	7	.5	.065	.188	.031	-037	-038
	Roller leaf spring	.070	10	.5	.080	.219	.031	-039	-040
	Hinge leaf	.010	3	.5	.060	.375	.160	-041	-042
	Hinge leaf	.020	3.5	.5	.093	.406	.156	-043	-044
F	Hinge leaf	.040	4	.5	.140	.515	.109	-045	-046
	Hinge leaf	.070	7	.75	.150	.515	.125	-047	-048
	Hinge roller leaf	.010	3	.5	.080	.281	.156	-049	-050
	Hinge roller leaf	.020	3.5	.5	.093	.312	.140	-051	-052
G	Hinge roller leaf	.040	4	.5	.131	.469	.109	-053	-054
	Hinge roller leaf	.070	7	.75	.150	.469	.125	-055	-056

1/ Characteristics may change ±25 percent after endurance test.
2/ For replacement cross reference see table VI.
3/ Nominal contact separation values are shown for information and classification purposes only. Measurement is not required.

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TABLE II. Electrical rating.

Load	28 V dc		115 V ac – 60 Hz	
	Current amperes	Number of operations	Current amperes	Number of operations
Resistive – sea level	2	100,000	15	100,000
50,000 ft	2	100,000	15	100,000
Inductive – sea level	1	100,000	15	100,000
50,000 ft	1	100,000	15	100,000
Lamp – sea level	1.5	100,000	1.5	100,000
50,000 ft	1.5	100,000	1.5	100,000

TABLE III. Qualification inspection group submission.

Examination or test	Basic switch	Other switch samples	Extent of approval
Qualification inspection table of MIL-PRF-8805.	M8805/1-038 (all sample units) M8805/1-034 (2 sample units) Table VI, group II.	2 sample units each of all other samples. 1. Visual and mechanical examination. 2. Operating characteristics	All

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TABLE IV. Direct current ratings. 1/
(Application Information)

Nominal contact separation in inches	Direct current	Noninductive				Inductive			
		Heater load		Lamp load		Motor, relay, or solenoid load			
						Sea level		50,000 feet	
		Normally open contact	Normally closed contact	Normally open contact	Normally closed contact	Normally open contact	Normally closed contact	Normally open contact	Normally closed contact
	(volts)	(amperes)	(amperes)	(amperes)	(amperes)	(amperes)	(amperes)	(amperes)	(amperes)
.010	6-8	15	20	1.5	3.0	8.0	8.0	7.0	7.0
	12-14	15	20	1.5	3.0	5.0	5.0	5.0	5.0
	24-30	2.0	2.0	1.5	2.0	1.0	1.0	1.0	1.0
	110-115	.4	.4	.4	.4	.03	.03	.02	.02
	220-230	.2	.2	.2	.2	.02	.02	.01	.01
.020	6-8	15	20	1.5	3.0	15	20	15	15
	12-14	15	20	1.5	3.0	10	10	8.0	8.0
	24-30	6.0	6.0	1.5	3.0	5.0	5.0	2.0	2.0
	110-115	.4	.4	.4	.4	.05	.05	.03	.03
	220-230	.2	.2	.2	.2	.03	.03	.02	.02
.040	6-8	15	20	1.5	3.0	15	20	15	15
	12-14	15	20	1.5	3.0	15	20	15	15
	24-30	10	10	1.5	3.0	10	10	5.0	5.0
	110-115	.6	.6	.6	.6	.1	.1	.05	.05
	220-230	.3	.3	.2	.3	.05	.05	.03	.03
.070	6-8	15	20	1.5	3.0	15	20	15	15
	12-14	15	20	1.5	3.0	15	20	15	15
	24-30	15	20	1.5	3.0	10	10	7.5	7.5
	110-115	.75	.75	.75	.75	.4	.4	.2	.2
	220-230	.3	.3	.3	.3	.2	.2	.1	.1

1/ Maximum inrush current 30 amperes for normally closed contacts and 15 amperes for normally open contacts

TABLE V Alternating current rating at 60 Hz. 1/
(Application information)

Alternating current	Heater load		Lamp load (tungsten)		Motor, relay, or solenoid load	
	Normally open contact	Normally closed contact	Normally open contact	Normally closed contact	Normally open contact	Normally closed contact
(Volts)	(amperes)	(amperes)	(amperes)	(amperes)	(amperes)	(amperes)
110-125	15.0	15.0	1.5	3.0	15.0	15.0
220-250	15.0	15.0	1.25	2.5	15.0	15.0
440-460 2/	3.0	3.0	.75	1.0	2.5	2.5
600 2/	2.0	2.0	.50	.8	2.0	2.0

1/ Maximum inrush current 30 amperes for normally closed contacts and 15 amperes for normally open contacts.
2/ These ratings apply only to sensitive switches having a nominal contact separation of .020, .040, or .070 inch. Nominal contact .010 is not rated for service at 440 volts or above.

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TABLE VI. Replacement cross reference.

PIN	Replaces JAN-S-63 unless otherwise noted	PIN	Replaces JAN-S-63 unless otherwise noted
M8805/1-001	SS04B20	M8805/1-029	SS04A40
M8805/1-002	SS04A20	M8805/1-030	SS04B40
M8805/1-003	MS35061 (SigC) SS01A20	M8805/1-031	SS04A70
M8805/1-004	MS35061 (SigC) SS01B20	M8805/1-032	SS04B70
M8805/1-005	SS01A40	M8805/1-033	SS05A10
M8805/1-006	SS01B40	M8805/1-034	SS05B10
M8805/1-007	MS35061 (SigC) SS01A70	M8805/1-035	MS35063(SigC) SS05A20
M8805/1-008	MS35061 (SigC) SS01B70, MS25026	M8805/1-036	MS35063(SigC) SS05B20
M8805/1-009	SS02A10	M8805/1-037	SS05A40
M8805/1-010	SS02B10	M8805/1-038	SS05B40
M8805/1-011	SS02A20	M8805/1-039	MS35063(SigC) SS05A70
M8805/1-012	SS02B20	M8805/1-040	MS35063(SigC) SS05B70
M8805/1-013	SS02A40	M8805/1-041	SS06A10
M8805/1-014	SS02B40	M8805/1-042	SS06B10
M8805/1-015	SS02A70	M8805/1-043	SS06A20
M8805/1-016	SS02B70	M8805/1-044	SS06B20
M8805/1-017	SS03A10	M8805/1-045	SS06A40
M8805/1-018	SS03B10	M8805/1-046	SS06B40
M8805/1-019	MS35062(SigC) SS03A20	M8805/1-047	SS06A70
M8805/1-020	MS35062(SigC) SS03B20	M8805/1-048	SS06B70
M8805/1-021	SS03A40	M8805/1-049	SS07A10
M8805/1-022	SS03B40	M8805/1-050	SS07B10
M8805/1-023	MS35062(SigC) SS03A70	M8805/1-051	SS07A20
M8805/1-024	MS35062(SigC) SS03B70	M8805/1-052	SS07B20
M8805/1-025	SS04A10	M8805/1-053	SS07A40
M8805/1-026	SS04B10	M8805/1-054	SS07B40
M8805/1-027	SS01B10	M8805/1-055	SS07A70
M8805/1-028	SS01A10	M8805/1-056	SS07B70

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Army - CR
Navy - EC
Air Force - 11
DLA - CC

Review activities:
Army - AR, MI
Navy - AS, MC, OS, SH
Air Force - 19, 99

Preparing activity:
DLA - CC

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